Project 1: Car Parking Tracker/Inventory REST API

API Tutorial

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ASE 230

What The Car Parking API does

A REST API for managing parked vehicles and the parking lot they are in.

10 endpoints total: all 10 use the parking database the 2 DELETE endpoints require Bearer token authentication.

Endpoints 1-5

#	Method	Endpoint	Auth
1	GET	/car	none
2	GET	/car/{plate}	none
3	DELETE	/car/{plate}	Bearer token
4	POST	/car	none
5	PUT	/car/{plate, color}	none

Endpoints 6-10

#	Method	Endpoint	Auth
6	GET	/lot/{make, color}	none
7	GET	/lot/	none
8	POST	/lot	none
9	PUT	/car/{lot, make}	none
10	DELETE	/lot/	Bearer token

Database

Database: `parkingdb`
Table: `parking` — car, plate, make, color, year, lot

For Database formato got to `code/schema.sql`.

GET /car

Request:

```
curl http://localhost:8080/car
```

gets all cars

Response:


```
[{"car": 1,"plate": "ABC-1234","make": "Toyota","color": "Red","year": 2020,"lot": 101}]
```

GET /car/{plate}

Request:

```
curl http://localhost:8080/car/ABC-1234
```

Response:


```
{"car": 1,"plate": "ABC-1234","make": "Toyota","color": "Red","year": 2020,"lot": 101}
```

DELETE /car/{plate} (secured)

Needs a Bearer token in the Authorization header to work.

```
curl -X DELETE "http://localhost:8080/car/ABC-1234" \
  -H "Authorization: Bearer john-demo-token"
```

Without a valid token, the server returns 401.

POST /car

Request:

```
curl -X POST "http://localhost:8080/car" \
  -H "Content-Type: application/json" \
  -d '{"plate": "XYZ-9999", "make": "Honda", "color": "Blue", "year": 2022, "lot": 102}'
```

Response:

PUT /car/{plate}/{color}

Needs a Bearer token in the Authorization header.

```
curl -X PUT "http://localhost:8080/car/XYZ-9999/Silver" \
  -H "Content-Type: application/json" \
  -d '{"plate": "XYZ-NEW"}'
```

Response:

```
{"success": true,"message": "Car updtated successfully"}
```

GET /lot/{lot}/{make}/{color}/{year}

Same pattern as students, on the `courses` table.

```
curl GET "http://localhost:8080/lot/102/Honda/Silver/2022"
```

Response:


```
[{"car": 2,"plate": "XYZ-NEW","make": "Honda","color": "Silver","year": 2022,"lot": 102}]
```

GET /lot

```
curl -X GET "http://localhost:8080/lot"
```

Response:

```
{"success": true,"data": [101, 102]}
```

POST /lot

```
curl -X POST "http://localhost:8080/lot" \
  -H "Content-Type: application/json" \
  -d '{"lot": 103}'
```

PUT /car/{lot}/{make}

```
curl -X PUT "http://localhost:8080/car/103/Honda"
```

Response:

```
{"success": true,"message": "Updated cars of make 'Honda' to lot 103"}
```

DELETE /lot/{lot}

```
curl -s -X DELETE "http://localhost:8080/lot/103" \
  -H "Authorization: Bearer admin-demo-token"
```

Response:

```
{"success": true,"message": "Lot 103 and associated cars deleted successfully"}
```

How authentication works

`auth.php` checks the `Authorization` header for `Bearer admin-demo-token` or other expected tokens.
If it doesn't match, the request is rejected with a 401 error before touching the database.

Testing

- `code/test_curl.sh` — command line tests for all 10 endpoints
- `code/test_client.html` — browser test page for all 10 endpoints

Next steps (Project 2)

The Parking API will later be rebuilt with Laravel, Eloquent ORM, and Docker. (definitly needed after the recent php exploits)

Thank you